Lohith Kovuri

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Summary

Software Engineer with production experience building backend systems, distributed infrastructure, and AI-powered services at scale. Shipped fault-tolerant event pipelines, RAG systems, and containerized APIs on AWS with measurable gains in reliability, latency, and operational visibility. Focused on systems correctness, failure handling, and observability. Experienced across the full delivery cycle from system design and implementation to incident response, observability instrumentation, and production ownership at a Fortune 100 retailer.

Skills

Languages: Python, Java, TypeScript, JavaScript, C++, SQL
Backend: FastAPI, Spring Boot, Node.js, REST APIs, microservices, distributed systems, event-driven architecture
Cloud & DevOps: AWS (ECS, S3, Lambda, SNS/SQS, CloudWatch), Azure (Databricks, Functions), Docker, Kubernetes, GitHub Actions, CI/CD
Data & AI: LangChain, RAG pipelines, FAISS, OpenAI API, prompt engineering, LLM evaluation (RAGAS)
Databases: PostgreSQL, MySQL, DynamoDB, Redis
Practices: Agile/Scrum, observability, distributed tracing, on-call, code review, performance profiling

Experience

Software Engineer | Nordstrom | Remote, USA **Oct 2025 – Present**

- Owned cloud-native AWS ingestion service persisting high-volume production events to S3 at **99.9% SLA**; handles burst traffic and partial failures with zero data loss.
- Rebuilt SNS/SQS consumer pipeline with idempotent message processing and exponential backoff, cutting MTTR on write failures by **40%** and reducing incident noise.
- Implemented ECS auto-scaling with CloudWatch alarms tuned to production traffic patterns; team began catching regressions before customer impact for the first time.
- Integrated OpenAI API into test generation workflow to auto-produce edge-case scripts, blocking an entire class of input validation bugs from reaching production.
- Led incident response on queue backlogs and S3 write failures; improved runbook coverage and observability metrics to accelerate future diagnosis.

Software Engineer | Rowan ACM Club | Remote, USA **Jun 2025 – Oct 2025**

- Architected conversational AI agent backend using LangChain and RAG pipelines; integrated live enterprise data sources with multi-turn context and hallucination guardrails.
- Enforced LLM endpoint reliability through input validation, output filtering, and privacy-safe session isolation across all inference paths.
- Built CI/CD pipeline with automated rollback gates, enabling zero-downtime releases throughout the project lifecycle.
- Owned sprint planning end-to-end: scoped features, decomposed work into time-estimated tasks, and surfaced blockers early—zero missed deadlines across all milestones.

Software Developer Intern | Roothoot Pvt Ltd | Hyderabad, India **Sep 2022 – Dec 2022**

- Profiled and optimized Java Spring Boot REST API endpoints, reducing P95 latency by **25%** through query tuning and caching improvements.
- Contributed to AWS deployment automation and CI/CD pipelines; collaborated across time zones to maintain sprint velocity.
- Caught multiple production-bound bugs through structured code review; improved code quality through feedback cycles with senior engineers.

Projects

Multi-Tenant RAG API | Python, FastAPI, LangChain, FAISS, Redis, PostgreSQL

- Built production-grade RAG API supporting **200+ concurrent tenants** with per-tenant FAISS namespace isolation at the embedding layer, not just at the API level.
- SHA-256 query fingerprint cache cut redundant OpenAI calls by **62%** and dropped average latency from **1.8s to 680ms**; Redis-backed token budgets prevent runaway API costs per tenant.
- Async RAGAS eval pipeline on live queries; automated alerts fire on faithfulness degradation before end-user impact.

Distributed Job Scheduler | Python, Redis, PostgreSQL, FastAPI, Docker, Prometheus

- Built distributed task scheduler with 6-state job machine; atomic claims via Redlock eliminated duplicate execution to zero under concurrent worker contention.
- Heartbeat-based stall detection auto-requeues failed jobs within **30s**; dead-letter queue captures permanent failures with full stack traces and one-click replay.
- `SKIP LOCKED` polling optimization brought p99 scheduling latency to **under 200ms** at 2,000 jobs/min.

Education

Rowan University | M.S. Data Science **2023 – 2025**
Dr. MGR Educational and Research Institute | B.Tech Information Technology **2019 – 2023**